

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

CHAPTER 10: SURFACE CHEMISTRY

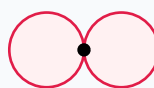
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Surface Chemistry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



p orbital (px)

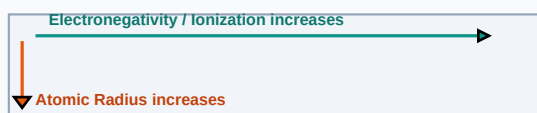
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Adsorption	The accumulation of molecular species at the surface rather than in the bulk of a solid or liquid.
Colloid	A heterogeneous system where one substance is dispersed (particles 1-1000nm) in another medium.
Tyndall Effect	The scattering of light rays by colloidal particles dispersed in a medium, illuminating the beam path.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

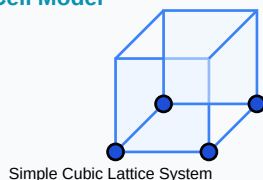
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Freundlich isotherm: $x/m = k \cdot P^{(1/n)}$ ($n > 1$)
- Electrophoretic velocity: balance of charge drag and electric field

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

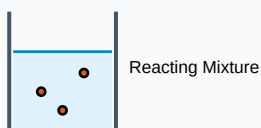
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Differentiate between physisorption and chemisorption.

Step-by-step Solution:

Physisorption involves weak van der Waals forces, is reversible, and lacks specificity. Chemisorption involves strong chemical bond formation, is irreversible, and is highly specific.

Titration Beaker & Solute Precipitation



HOTS (Higher Order Thinking Skills) Challenge

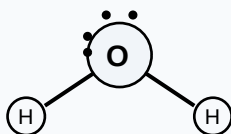
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain what homogeneous and heterogeneous catalysis are, detailing the adsorption theory of heterogeneous catalysts.

Analytical Solution Strategy:

(1) Homogeneous: catalyst and reactants in same phase. (2) Heterogeneous: different phases. Adsorption theory steps: (a) Diffusion of reactants to catalyst surface, (b) Adsorption, (c) Chemical reaction (forming activated complex), (d) Desorption of products, (e) Diffusion of products away.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

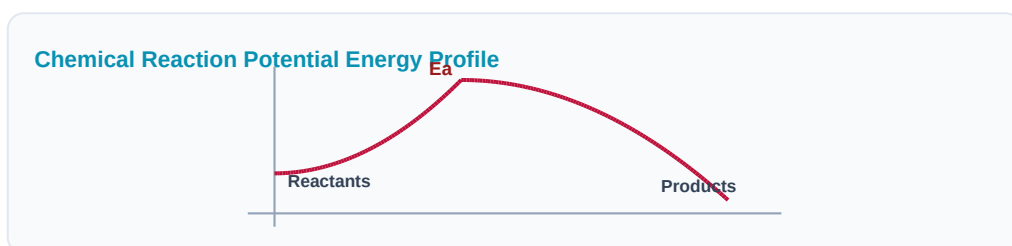
Q1 (1-Mark): What is Freundlich adsorption isotherm? Write the mathematical expression.

Q2 (2-Mark): Classify colloids based on dispersed phase and dispersion medium.

Q3 (2-Mark): Describe the purification of colloids by Dialysis.

Q4 (5-Mark): State Hardy-Schulze rule for coagulation.

Q5 (5-Mark): Explain emulsion types (oil-in-water and water-in-oil).



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Gas Masks

Activated charcoal filters adsorb toxic gas molecules, keeping air passages clean.

Application Case: Medicine

Intravenous drug emulsions are formulated as colloids to deliver lipophilic medicines directly.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

